# CHEMICALS IN COSMETICS

## Introduction

***The aim of the project is to identify the amount of chemical substances present in a Cosmetic product.The California Safe Cosmetics Program (CSCP), administered by the California Department of Public Health (CDPH), is a vital initiative designed to protect public health by monitoring and disclosing hazardous ingredients in cosmetic products sold in California. The program, established under the California Safe Cosmetics Act, requires cosmetic manufacturers, packers, and distributors to report products containing ingredients that are known or suspected to cause cancer, birth defects, or other developmental or reproductive harm.*** ¶

## Goal of the Project

```
The goal of this project is to analyze and assess the data reported t
o the California Safe Cosmetics Program (CSCP) to better understand t
he presence and prevalence of hazardous ingredients in cosmetic produ
cts sold in California. By examining this data, the project aims to:
```

1. Identify trends in the use of chemicals known or suspected to cause cancer, birth defects, or other developmental or reproductive harm within the cosmetic industry.
2. Evaluate the compliance of manufacturers with the reporting requirements set by the California Safe Cosmetics Act, and assess the completeness of the data collected by the CSCP.
3. Increase awareness about the potential health risks associated with certain cosmetic ingredients, and provide insights into how these chemicals are distributed across different product categories and brands.
4. Highlight gaps in the reporting process, including missing data or products that may not be included, and suggest ways to improve transparency and safety within the cosmetics industry.
5. Support consumer education by presenting the data in a user-friendly format, helping consumers make more informed decisions about the personal care products they use.

In [3]:
```python
import numpy as np
import pandas as pd
```

## Load Dataset

- This dataset is from the California Safe Cosmetics Program (CSCP) in the California Department of Public Health
- The primary purpose of the CSCP is to collect information on hazardous and potentially hazardous ingredients in cosmetic products sold in California and to make this information available to the public.

- For all cosmetic products sold in California, the California Safe Cosmetics Act requires the manufacturer, packer, and/or distributor named on the product label to provide to the CSCP a list of all cosmetic products that contain any ingredients known or suspected to cause cancer, birth defects, or other developmental or reproductive harm.
- The CSCP maintains a list of “reportable” ingredients. Companies with reportable ingredients in their products must submit information to the California Safe Cosmetics Program.
- The data table consists of: label names of cosmetic/personal care products, company/manufacturer names, product brand names, product categories, Chemical Abstracts Service registry numbers (CAS#) of the reported chemical ingredients, names of reported chemical ingredients, the number of reported chemicals for each product, and dates of reporting, product discontinuation or reformulation if applicable.

In [61]:
```
df=pd.read_csv("C:/Users/user/Downloads/cscpopendata.csv")
df
```

Out[61]:

| | CDPHId | ProductName | CSFId | CSF | CompanyId | CompanyName | BrandName | Prir |
|---|---|---|---|---|---|---|---|---|
| **0** | 2 | ULTRA COLOR RICH EXTRA PLUMP LIPSTICK-ALL SHADES | NaN | NaN | 4 | New Avon LLC | AVON | |
| **1** | 3 | Glover's Medicated Shampoo | NaN | NaN | 338 | J. Strickland & Co. | Glover's | |
| **2** | 3 | Glover's Medicated Shampoo | NaN | NaN | 338 | J. Strickland & Co. | Glover's | |
| **3** | 4 | PRECISION GLIMMER EYE LINER-ALL SHADES ◆ | NaN | NaN | 4 | New Avon LLC | AVON | |
| **4** | 5 | AVON BRILLIANT SHINE LIP GLOSS-ALL SHADES ◆ | NaN | NaN | 4 | New Avon LLC | AVON | |
| **...** | ... | ... | ... | ... | ... | ... | ... | |
| **114630** | 41523 | HYDRA-LIP TRANSLUCENT COLOR LIPSTICK | 65001.0 | Rosa Soft | 1259 | Yanbal USA, Inc | YANBAL | |
| **114631** | 41523 | HYDRA-LIP TRANSLUCENT COLOR LIPSTICK | 65002.0 | Malva Spirit | 1259 | Yanbal USA, Inc | YANBAL | |
| **114632** | 41523 | HYDRA-LIP TRANSLUCENT COLOR LIPSTICK | 65003.0 | Rojo Fashion | 1259 | Yanbal USA, Inc | YANBAL | |
| **114633** | 41523 | HYDRA-LIP TRANSLUCENT COLOR LIPSTICK | 65004.0 | Terra Mystic | 1259 | Yanbal USA, Inc | YANBAL | |
| **114634** | 41524 | OLD SPICE GENTLEMENS BLEND ALOE AND WILD SAGE ... | NaN | NaN | 86 | The Procter & Gamble Company | Old Spice | |

114635 rows × 22 columns

In [ ]:

# Identifying Data Structure

In [4]:
```
df.head()
```

Out[4]:

| | CDPHId | ProductName | CSFId | CSF | CompanyId | CompanyName | BrandName | PrimaryCategor |
|---|---|---|---|---|---|---|---|---|
| **0** | 2 | ULTRA COLOR RICH EXTRA PLUMP LIPSTICK-ALL SHADES | NaN | NaN | 4 | New Avon LLC | AVON | |
| **1** | 3 | Glover's Medicated Shampoo | NaN | NaN | 338 | J. Strickland & Co. | Glover's | |
| **2** | 3 | Glover's Medicated Shampoo | NaN | NaN | 338 | J. Strickland & Co. | Glover's | |
| **3** | 4 | PRECISION GLIMMER EYE LINER-ALL SHADES  | NaN | NaN | 4 | New Avon LLC | AVON | |
| **4** | 5 | AVON BRILLIANT SHINE LIP GLOSS-ALL SHADES  | NaN | NaN | 4 | New Avon LLC | AVON | |

5 rows × 22 columns

In [5]:
```
df.tail()
```

Out[5]:

| | CDPHId | ProductName | CSFId | CSF | CompanyId | CompanyName | BrandName | Prir |
|---|---|---|---|---|---|---|---|---|
| **114630** | 41523 | HYDRA-LIP TRANSLUCENT COLOR LIPSTICK | 65001.0 | Rosa Soft | 1259 | Yanbal USA, Inc | YANBAL | |
| **114631** | 41523 | HYDRA-LIP TRANSLUCENT COLOR LIPSTICK | 65002.0 | Malva Spirit | 1259 | Yanbal USA, Inc | YANBAL | |
| **114632** | 41523 | HYDRA-LIP TRANSLUCENT COLOR LIPSTICK | 65003.0 | Rojo Fashion | 1259 | Yanbal USA, Inc | YANBAL | |
| **114633** | 41523 | HYDRA-LIP TRANSLUCENT COLOR LIPSTICK | 65004.0 | Terra Mystic | 1259 | Yanbal USA, Inc | YANBAL | |
| **114634** | 41524 | OLD SPICE GENTLEMENS BLEND ALOE AND WILD SAGE ... | NaN | NaN | 86 | The Procter & Gamble Company | Old Spice | |

5 rows × 22 columns

In [6]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 114635 entries, 0 to 114634
Data columns (total 22 columns):
 #   Column                  Non-Null Count   Dtype  
---  ------                  --------------   -----  
 0   CDPHId                  114635 non-null  int64  
 1   ProductName             114635 non-null  object 
 2   CSFId                   80662 non-null   float64
 3   CSF                     80237 non-null   object 
 4   CompanyId               114635 non-null  int64  
 5   CompanyName             114635 non-null  object 
 6   BrandName               114408 non-null  object 
 7   PrimaryCategoryId       114635 non-null  int64  
 8   PrimaryCategory         114635 non-null  object 
 9   SubCategoryId           114635 non-null  int64  
 10  SubCategory             114635 non-null  object 
 11  CasId                   114635 non-null  int64  
 12  CasNumber               108159 non-null  object 
 13  ChemicalId              114635 non-null  int64  
 14  ChemicalName            114635 non-null  object 
 15  InitialDateReported     114635 non-null  object 
 16  MostRecentDateReported  114635 non-null  object 
 17  DiscontinuedDate        12920 non-null   object 
 18  ChemicalCreatedAt       114635 non-null  object 
 19  ChemicalUpdatedAt       114635 non-null  object 
 20  ChemicalDateRemoved     2985 non-null    object 
 21  ChemicalCount           114635 non-null  int64  
dtypes: float64(1), int64(7), object(14)
memory usage: 19.2+ MB
```

In [8]:
```
df.columns
```

Out[8]:
```
Index(['CDPHId', 'ProductName', 'CSFId', 'CSF', 'CompanyId', 'CompanyName',
       'BrandName', 'PrimaryCategoryId', 'PrimaryCategory', 'SubCategoryId',
       'SubCategory', 'CasId', 'CasNumber', 'ChemicalId', 'ChemicalName',
       'InitialDateReported', 'MostRecentDateReported', 'DiscontinuedDate',
       'ChemicalCreatedAt', 'ChemicalUpdatedAt', 'ChemicalDateRemoved',
       'ChemicalCount'],
      dtype='object')
```

In [9]:
```
df.shape
```

Out[9]:
```
(114635, 22)
```

In [11]:
```
df.describe()
```

Out[11]:

| | CDPHId | CSFId | CompanyId | PrimaryCategoryId | SubCategoryId | C |
|---|---|---|---|---|---|---|
| **count** | 114635.000000 | 80662.000000 | 114635.000000 | 114635.000000 | 114635.000000 | 114635.00 |
| **mean** | 20304.858987 | 32608.658377 | 450.641532 | 51.076294 | 66.819252 | 674.09 |
| **std** | 12489.052554 | 19089.443910 | 409.533093 | 20.474341 | 35.822097 | 149.21 |
| **min** | 2.000000 | 1.000000 | 4.000000 | 1.000000 | 3.000000 | 2.00 |
| **25%** | 8717.000000 | 15789.000000 | 86.000000 | 44.000000 | 48.000000 | 656.00 |
| **50%** | 20895.000000 | 32541.000000 | 297.000000 | 44.000000 | 52.000000 | 656.00 |
| **75%** | 31338.500000 | 48717.750000 | 798.000000 | 59.000000 | 65.000000 | 656.00 |
| **max** | 41524.000000 | 65009.000000 | 1391.000000 | 111.000000 | 172.000000 | 1242.00 |

# Identifying the Null Values

In [13]:
```
df.isnull().sum()
```

Out[13]:
```
CDPHId                      0
ProductName                 0
CSFId                   33973
CSF                     34398
CompanyId                   0
CompanyName                 0
BrandName                 227
PrimaryCategoryId           0
PrimaryCategory             0
SubCategoryId               0
SubCategory                 0
CasId                       0
CasNumber                6476
ChemicalId                  0
ChemicalName                0
InitialDateReported         0
MostRecentDateReported      0
DiscontinuedDate       101715
ChemicalCreatedAt           0
ChemicalUpdatedAt           0
ChemicalDateRemoved    111650
ChemicalCount               0
dtype: int64
```

# Treating Null Values

In [93]:
```
df.drop(columns=["CSFId", "CSF", "CasNumber", "ChemicalDateRemoved","Discontin
df
```

Out[93]:

| | CDPHId | ProductName | CompanyName | BrandName | PrimaryCategoryId | SubCategor |
|---|---|---|---|---|---|---|
| **0** | 2 | ULTRA COLOR RICH EXTRA PLUMP LIPSTICK-ALL SHADES | New Avon LLC | AVON | 44 | Lip Color Lipsticks, Liners and Pencil |
| **1** | 3 | Glover's Medicated Shampoo | J. Strickland & Co. | Glover's | 18 | Hair Shampoo (making cosmetic claim |
| **2** | 3 | Glover's Medicated Shampoo | J. Strickland & Co. | Glover's | 18 | Hair Shampoo (making cosmetic claim |
| **3** | 4 | PRECISION GLIMMER EYE LINER-ALL SHADES ◆ | New Avon LLC | AVON | 44 | Eyeliner/Eyebrow Pencil |
| **4** | 5 | AVON BRILLIANT SHINE LIP GLOSS-ALL SHADES ◆ | New Avon LLC | AVON | 44 | Lip Gloss/Shine |
| **...** | ... | ... | ... | ... | ... | ... |
| **114630** | 41523 | HYDRA-LIP TRANSLUCENT COLOR LIPSTICK | Yanbal USA, Inc | YANBAL | 44 | Lip Color Lipsticks, Liners and Pencil |
| **114631** | 41523 | HYDRA-LIP TRANSLUCENT COLOR LIPSTICK | Yanbal USA, Inc | YANBAL | 44 | Lip Color Lipsticks, Liners and Pencil |
| **114632** | 41523 | HYDRA-LIP TRANSLUCENT COLOR LIPSTICK | Yanbal USA, Inc | YANBAL | 44 | Lip Color Lipsticks, Liners and Pencil |
| **114633** | 41523 | HYDRA-LIP TRANSLUCENT COLOR LIPSTICK | Yanbal USA, Inc | YANBAL | 44 | Lip Color Lipsticks, Liners and Pencil |
| **114634** | 41524 | OLD SPICE GENTLEMENS BLEND ALOE AND WILD SAGE ... | The Procter & Gamble Company | Old Spice | 6 | Body Washes and Soaps |

114635 rows × 14 columns

In [94]:
```
df.isnull().sum()
```

Out[94]:
```
CDPHId                     0
ProductName                0
CompanyName                0
BrandName                227
PrimaryCategoryId          0
SubCategory                0
CasId                      0
ChemicalId                 0
ChemicalName               0
InitialDateReported        0
MostRecentDateReported     0
ChemicalCreatedAt          0
ChemicalUpdatedAt          0
ChemicalCount              0
dtype: int64
```

In [95]:
```
df['BrandName'].replace('Greenbrier International, Inc.', 'Essentials', inplac
df["BrandName"].fillna("Merle Norman Cosmetics",inplace=True)
```

In [96]:
```
df.isnull().sum()
```

Out[96]:
```
CDPHId                   0
ProductName              0
CompanyName              0
BrandName                0
PrimaryCategoryId        0
SubCategory              0
CasId                    0
ChemicalId               0
ChemicalName             0
InitialDateReported      0
MostRecentDateReported   0
ChemicalCreatedAt        0
ChemicalUpdatedAt        0
ChemicalCount            0
dtype: int64
```

# Handling Duplicate Values

In [97]:
```
df.duplicated().sum()
```

Out[97]:
```
50402
```

In [98]:
```
data=df.drop_duplicates()
```

In [99]:
```
data
```

Out[99]:

| | CDPHId | ProductName | CompanyName | BrandName | PrimaryCategoryId | SubCategor |
|---|---|---|---|---|---|---|
| **0** | 2 | ULTRA COLOR RICH EXTRA PLUMP LIPSTICK-ALL SHADES | New Avon LLC | AVON | 44 | Lip Color Lipsticks, Liners and Pencil |
| **1** | 3 | Glover's Medicated Shampoo | J. Strickland & Co. | Glover's | 18 | Hair Shampoo (making cosmetic claim |
| **2** | 3 | Glover's Medicated Shampoo | J. Strickland & Co. | Glover's | 18 | Hair Shampoo (making cosmetic claim |
| **3** | 4 | PRECISION GLIMMER EYE LINER-ALL SHADES ⬚ | New Avon LLC | AVON | 44 | Eyeliner/Eyebrow Pencil |
| **4** | 5 | AVON BRILLIANT SHINE LIP GLOSS-ALL SHADES ⬚ | New Avon LLC | AVON | 44 | Lip Gloss/Shin |
| **...** | ... | ... | ... | ... | ... | ... |
| **114630** | 41523 | HYDRA-LIP TRANSLUCENT COLOR LIPSTICK | Yanbal USA, Inc | YANBAL | 44 | Lip Color Lipsticks, Liners and Pencil |
| **114631** | 41523 | HYDRA-LIP TRANSLUCENT COLOR LIPSTICK | Yanbal USA, Inc | YANBAL | 44 | Lip Color Lipsticks, Liners and Pencil |
| **114632** | 41523 | HYDRA-LIP TRANSLUCENT COLOR LIPSTICK | Yanbal USA, Inc | YANBAL | 44 | Lip Color Lipsticks, Liners and Pencil |
| **114633** | 41523 | HYDRA-LIP TRANSLUCENT COLOR LIPSTICK | Yanbal USA, Inc | YANBAL | 44 | Lip Color Lipsticks, Liners and Pencil |
| **114634** | 41524 | OLD SPICE GENTLEMENS BLEND ALOE AND WILD SAGE ... | The Procter & Gamble Company | Old Spice | 6 | Body Washes and Soaps |

64233 rows × 14 columns

# Outlier Detection

In [ ]:
```
import seaborn as sns
import matplotlib.pyplot as plt
```

In [125]:
```
num_data=data.select_dtypes(include=["number"])
num_data
```

Out[125]:

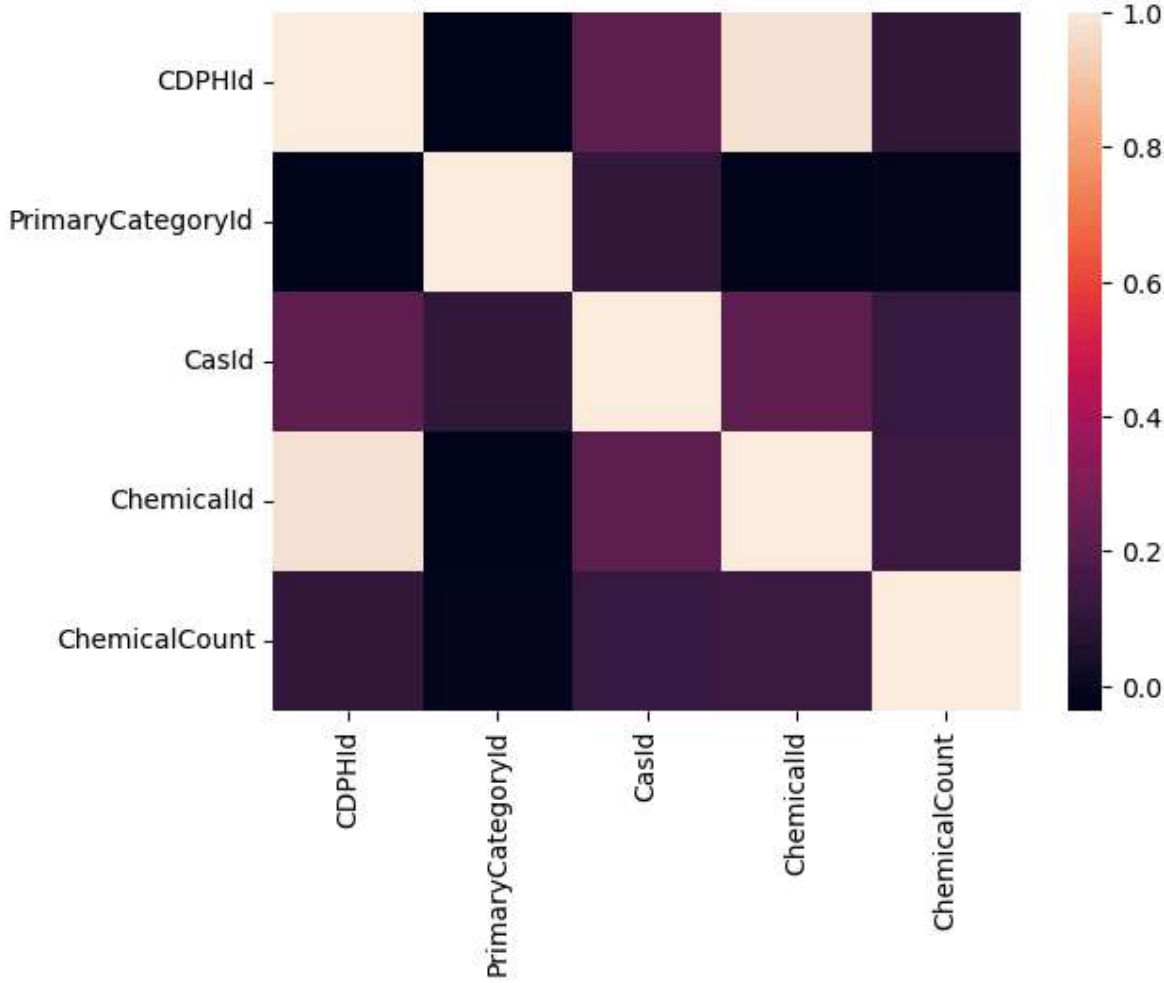
| | CDPHId | ProductName | CompanyName | BrandName | PrimaryCategoryId | SubCategory | Ca |
|---|---|---|---|---|---|---|---|
| **0** | 2 | 30981 | 372 | 82 | 44 | 40 | |
| **1** | 3 | 13104 | 252 | 1023 | 18 | 32 | |
| **2** | 3 | 13104 | 252 | 1023 | 18 | 32 | |
| **3** | 4 | 22843 | 372 | 82 | 44 | 19 | |
| **4** | 5 | 1328 | 372 | 82 | 44 | 41 | |
| **...** | ... | ... | ... | ... | ... | ... | |
| **114630** | 41523 | 13606 | 579 | 2578 | 44 | 40 | |
| **114631** | 41523 | 13606 | 579 | 2578 | 44 | 40 | |
| **114632** | 41523 | 13606 | 579 | 2578 | 44 | 40 | |
| **114633** | 41523 | 13606 | 579 | 2578 | 44 | 40 | |
| **114634** | 41524 | 21965 | 527 | 1729 | 6 | 11 | |

64233 rows × 14 columns

In [100]:
```
s=data.select_dtypes(include="number").corr()
```

In [101]:
```
sns.heatmap(s)
```

Out[101]:
```
<Axes: >
```




# ## Encoding

In [102]:
```
object_columns = data.select_dtypes(include='object').columns
object_columns
```

Out[102]:
```
Index(['ProductName', 'CompanyName', 'BrandName', 'SubCategory',
       'ChemicalName', 'InitialDateReported', 'MostRecentDateReported',
       'ChemicalCreatedAt', 'ChemicalUpdatedAt'],
      dtype='object')
```

In [103]:
```
from sklearn.preprocessing import OneHotEncoder,LabelEncoder
```

In [104]:
```
encoder=LabelEncoder()
```

In [118]:
```
data=data.copy()
data['ProductName']=encoder.fit_transform(data['ProductName'])
data['CompanyName']=encoder.fit_transform(data['CompanyName'])
data['BrandName']=encoder.fit_transform(data['BrandName'])
data['SubCategory']=encoder.fit_transform(data['SubCategory'])
data['ChemicalName']=encoder.fit_transform(data['ChemicalName'])
data['InitialDateReported']=encoder.fit_transform(data['InitialDateReported'])
data[ 'MostRecentDateReported']=encoder.fit_transform(data['MostRecentDateRepo
data['ChemicalCreatedAt']=encoder.fit_transform(data['ChemicalCreatedAt'])
data['ChemicalUpdatedAt']=encoder.fit_transform(data['ChemicalUpdatedAt'])
```

In [119]:
```
data
```

Out[119]:

| | CDPHId | ProductName | CompanyName | BrandName | PrimaryCategoryId | SubCategory | Ca |
|---|---|---|---|---|---|---|---|
| **0** | 2 | 30981 | 372 | 82 | 44 | 40 | |
| **1** | 3 | 13104 | 252 | 1023 | 18 | 32 | |
| **2** | 3 | 13104 | 252 | 1023 | 18 | 32 | |
| **3** | 4 | 22843 | 372 | 82 | 44 | 19 | |
| **4** | 5 | 1328 | 372 | 82 | 44 | 41 | |
| **...** | ... | ... | ... | ... | ... | ... | |
| **114630** | 41523 | 13606 | 579 | 2578 | 44 | 40 | |
| **114631** | 41523 | 13606 | 579 | 2578 | 44 | 40 | |
| **114632** | 41523 | 13606 | 579 | 2578 | 44 | 40 | |
| **114633** | 41523 | 13606 | 579 | 2578 | 44 | 40 | |
| **114634** | 41524 | 21965 | 527 | 1729 | 6 | 11 | |

64233 rows × 14 columns

# ## Standardizing Features

In [120]:
```
from sklearn.preprocessing import StandardScaler
```

In [121]:
```
scaler = StandardScaler()
```

In [128]:
```
numerical_columns=data.select_dtypes(include="number").columns
```

In [129]:
```
data[numerical_columns] = scaler.fit_transform(data[numerical_columns])
data
```

Out[129]:

| | CDPHId | ProductName | CompanyName | BrandName | PrimaryCategoryId | SubCategory |
|---|---|---|---|---|---|---|
| **0** | -1.840546 | 1.478132 | 0.333704 | -1.683351 | -0.352986 | 0.072727 |
| **1** | -1.840462 | -0.404559 | -0.319358 | -0.458028 | -1.541072 | -0.266873 |
| **2** | -1.840462 | -0.404559 | -0.319358 | -0.458028 | -1.541072 | -0.266873 |
| **3** | -1.840378 | 0.621090 | 0.333704 | -1.683351 | -0.352986 | -0.818724 |
| **4** | -1.840294 | -1.644732 | 0.333704 | -1.683351 | -0.352986 | 0.115178 |
| **...** | ... | ... | ... | ... | ... | ... |
| **114630** | 1.638686 | -0.351692 | 1.460235 | 1.566814 | -0.352986 | 0.072727 |
| **114631** | 1.638686 | -0.351692 | 1.460235 | 1.566814 | -0.352986 | 0.072727 |
| **114632** | 1.638686 | -0.351692 | 1.460235 | 1.566814 | -0.352986 | 0.072727 |
| **114633** | 1.638686 | -0.351692 | 1.460235 | 1.566814 | -0.352986 | 0.072727 |
| **114634** | 1.638770 | 0.528625 | 1.177242 | 0.461289 | -2.089419 | -1.158325 |

64233 rows × 14 columns

In [ ]: